

AI PORTFOLIO

AI-Governed Infrastructure Automation

Six cloud, network, private-cloud, and telco automation playbooks with deterministic controls, human approval boundaries, and AI-assisted remediation.

Live Demo

GitHub Actions

Python · Ansible



Automation is useful only when it is governed



Infrastructure drift hides risk

Cloud, network, and platform changes happen across many systems. Without consistent checks, small drift can become outages, security exposure, or rework.



AI alone is not authorization

A model recommendation is not enough to change production infrastructure. Auto-remediation needs policy, blast-radius checks, rollback, and audit evidence.



Manual approval slows low-risk fixes

Escalating every issue protects safety, but it also leaves routine, reversible issues in manual queues and reduces the value of automation.

Let AI recommend — let policy decide

Model-only automation

Fast, but unsafe. A confident model can still be wrong, incomplete, or missing operational context.

Rejected

Manual-only operations

Safe, but inefficient. If every low-risk fix waits for a human, automation becomes a dashboard.

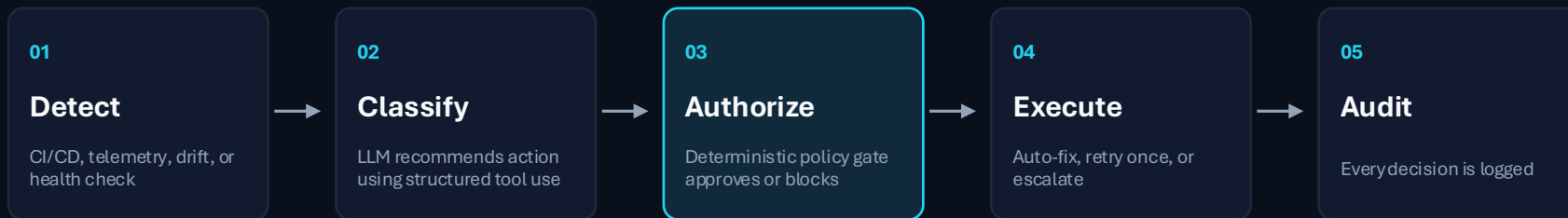
Low value

Governed autonomy

The model classifies the event. Deterministic controls decide whether to auto-fix, retry once, or escalate.

This project

One control pattern applied across six infrastructure domains



Key principle: the LLM recommends an action; deterministic policy, risk classification, precedent, rollback, and change-window checks authorize execution.

Cloud, network, private-cloud, and telco automation scenarios

01 VeloCloud SD-WAN

Branch onboarding, ZTP, profile policy

02 Multi-Cloud VPC

AWS · GCP · Azure shared CIDR pool

03 OpenStack VM

Heat stack + post-boot service config

04 VCF Catalog

Self-service blueprints, quota, cost tags

05 DNS & IPAM

NetBox source of truth + DNS automation

06 Network Compliance

NAPALM backup, gNMI drift, NETCONF fix path

The policy gate is separate from the model

Execution requires all of these:

- ✓ Action is on allowlist
- ✓ Resource risk is low
- ✓ Precedent is verified
- ✓ Change window is open
- ✓ Rollback path exists
- ✓ No repeated failure



Why it matters

A confident classification is not the same as a safe infrastructure change. The gate keeps authorization in deterministic code and policy, not inside a prompt.

When any required condition fails, the system escalates instead of guessing.

Auto-fix, retry once, or escalate

1

Auto-fix

Allowed only for known, low-risk, reversible changes with approved actions and rollback evidence.

2

Retry once

Used for transient failures where one bounded retry is safer than repeated attempts.

3

Escalate

Default path for novel events, high-risk resources, unclear precedent, or missing rollback.

What a real deployment would need beyond the demo

✓ **Keep secrets safe**

Use managed secrets and short-lived credentials, never hard-coded tokens.

✓ **Prevent duplicate actions**

Use locking and idempotency so two jobs do not make the same change twice.

✓ **Record every decision**

Keep an audit trail of what was recommended, approved, executed, or escalated.

✓ **Make changes reversible**

Require rollback steps before any auto-fix can run.

✓ **Protect the model input**

Treat logs and telemetry as untrusted data; sanitize before sending to the model.

✓ **Fail safely**

If the agent, API, or policy check fails, stop and escalate.

Every decision shows why it was allowed or blocked

Decision Evidence & Policy Evaluation

VPC-TAG-042	AUTO-FIX	cost-center tag missing; action allowlisted
SVC-BOOT-017	RETRY ONCE	known transient boot timing issue
SNMP-DRIFT-004	ESCALATE	security-sensitive configuration drift



What gets logged

- trigger event
- recommended action
- policy checks passed/failed
- human escalation reason
- execution result

The workflow at a glance

6

Infrastructure domains

3

SD-WAN · cloud · OpenStack ·
VCF · IPAM · compliance

Decision paths

1

auto-fix · retry once · escalate

Shared agent pattern

common control loop, domain-
specific tools

100%

Governed decisions

CI/CD

policy gate before execution

Validation workflow

HITL

repository checks on push and
PR

Approval boundary

risky or novel events are
escalated

TRY IT

Run a full infrastructure scenario in the browser

- ✓ **Governed by design** The model recommends, deterministic policy authorizes, and risky actions escalate.
- ✓ **Architecture-first** Each playbook includes problem, methodology, HLD, LLD, ADRs, and trade-offs.
- ✓ **Ready for review** Live demo, GitHub repo, validation workflow, and reference implementation are public.

Live demo · <https://danvzla.github.io/ai-governed-infrastructure-automation/>

github.com/danvzla/ai-governed-infrastructure-automation